

Akhilesh Pothuri

(+1)980-371-0953 | pothuriakhilesh@yahoo.com | [linkedin.com/in/akhilesh-pothuri](https://www.linkedin.com/in/akhilesh-pothuri) | github.com/AKhileshPothuri | medium.com/@pothuriakhilesh

EXPERIENCE

Verizon

Remote, United States

Machine Learning Engineer - GenAI

February 2024 - Present

- Leading the architecture of **Q-Agent**, a general-purpose AI agent platform (7 microservices) for natural-language data querying at enterprise scale. Designed the system as a unified chatbot with pluggable agents, MCP-based tool aggregation, and autonomous task routing across BigQuery, PostgreSQL, and GCS.
- Built a **LangGraph ReAct** routing layer with MultiServerMCPClient connecting 3 FastMCP servers (query orchestration, onboarding,retrieval); agents
- Engineered a **hybrid search** pipeline (BM25 full-text + pgvector cosine similarity) with Reciprocal Rank Fusion (RRF k=60),position-aware score blending, and Vertex AI text-embedding-005 for semantic retrieval over chunked skill metadata and knowledge documents.
- Designed an automated data onboarding pipeline, BigQuery INFORMATION SCHEMA → SKILL.md,tables.md generation → GCS versioned snapshots → markdown-aware chunking (heading-scored breaks, SHA256 dedup) → **Vertex AI** embedding → pgvector indexing.
- **Architected QVerse**, an enterprise-grade **Self-Serve Conversational BI** platform utilizing a hierarchical **Planner → Orchestrator → Reflector** pattern to process **10,000+ daily queries** with sub-second latency; built an AI-powered Metadata Builder for automated DB schema onboarding with human-in-the-loop validation.
- Developed **Talk2Report**, a ReAct-based agentic workflow with pgvector semantic search and citation tracking for verifiable RAG responses; implemented an Intelligent Routing Layer using **LangGraph** for intent classification, coreference resolution, and progressive prompt disclosure.
- Built ThinkForge, a query optimization engine using **hybrid semantic caching** and execution-based **VES metrics**, achieving a 60% cache hit rate and reducing LLM API costs by 40%.
- Led **reward-based fine-tuning (SFT + GRPO)** of Gemma and Llama models on distributed NVIDIA H100 clusters, utilizing **LoRA adapters** and custom programmatic reward functions to boost accuracy by 25%.
- Optimized **NLQ-to-SQL** pipelines using structured prompt orchestration and **Chain-of-Thought (CoT)** reasoning, improving query accuracy by 35% as validated through systematic A/B testing.
- Developed a multimodal troubleshooting pipeline using Gemini 2.0 Flash and Whisper to extract structured diagnostic steps from WebEx recordings via frame detection and audio-visual correlation.
- Productionized and orchestrated complex multi-agent sequences using FastAPI and **Docker** on **Kubernetes**; implemented **Model Context Protocol (MCP)** and **A2A** communication for autonomous task decomposition and fault-tolerant 'call-execute-reason' loops across 30+ domain-specific tools.
- Established a rigorous **LLM observability** and evaluation framework using **RAGAS** and **VES metrics** to monitor faithfulness, latency, and hallucination detection, optimizing inference to handle 10,000+ daily queries.

UNC Charlotte Football

Charlotte, NC

Data Scientist

August 2023 - December 2023

- Built an automated live tracking system using **Convolutional Neural Networks (CNN)** & advanced image processing algorithms (**computer vision**) for real-time player & jersey number recognition, saving 5 hours daily.
- Employed advanced statistical models, including **regression** and **clustering** to assess player strengths, weaknesses, and overall contribution to the team's performance.
- Utilized **SQL** for complex data manipulation tasks, including merging, aggregation, & **KPI** calculation for players.

Hewlett Packard R&D

Bangalore, India

Data Scientist

February 2020 - July 2022

- Led POC project to analyze global call volume data using **time series forecasting**, achieving a 28% mape boost.
- Conducted **operational analysis** to enhance risk mitigation & resource allocation, leveraging data-driven insights for informed decision-making. Employed predictive analytics for strategic optimization of supply chain resources.
- Performed ad-hoc data analysis in Python, fabricated a **data warehouse** & optimised SQL queries in **BigQuery**
- Developed and maintained financial models for forecasting, **budget planning**, and scenario analysis, leading to 15% improvement in budget accuracy and 20% reduction in forecasting errors.
- Built a spare part replacement **recommendation engine** with **natural language processing (NLP)** such as Transformers, reducing customer wait time to 15s enhancing user experience.
- Fine-tuned a **PyTorch based LSTM** model within the recommendation engine, optimizing its performance in understanding sequential patterns and improving the accuracy of personalized recommendations.

- Applied Dimensionality reduction & One-Class SVM for **anomaly detection** in complex, high-dimensional data.
- Spearheaded **end-to-end** development and deployment of **Demand Forecasting** projects, employing advanced techniques like ABC/XYZ classification and ADI/CV square-based classification for precise analysis.
- Executed comprehensive analysis on demand forecasting data, leveraging statistics and machine learning techniques to optimize **inventory management**.
- Employed monthly snapshots of spare parts demand for exploratory data analysis (EDA), **feature engineering** & modeling in Python & R, yielding an anticipated **\$26M ROI** over 3 years, optimizing supply chain efficiency.
- Automated **ETL** processes, model training, & evaluation via scalable **end-to-end** data infrastructure pipelines in **Azure Databricks** & **Azure Data Factory**, leveraging **Spark** clusters, saving 120 labor hours monthly.
- Delivered **actionable insights** to cross-functional teams, stakeholders through interactive **PowerBI** dashboards

SKILLS

- **Agentic AI & Orchestration:** ReAct, Chain-of-Thought (CoT), Planner-Orchestrator-Reflector patterns; LangGraph, LangChain, Model Context Protocol (MCP); Intent Routing, Multi-agent Task Decomposition, and Autonomous Workflows.
- **LLM Fine-Tuning & Training:** Supervised Fine-Tuning (SFT), PEFT (LoRA, QLoRA), Reinforcement Learning from Human Feedback (RLHF); Proximal Policy Optimization (PPO, GRPO); Distributed training on NVIDIA H100 clusters.
- **RAG & Retrieval Systems:** Retrieval-Augmented Generation (RAG), Semantic & Hybrid Search, pgvector, ChromaDB, FAISS; Layout-aware parsing, Metadata filtering, and Citation grounding.
- **LLM Evaluation & Observability:** Evaluation Frameworks (RAGAS, automated benchmarking suites); Metrics for Accuracy, Relevance, Hallucination Detection, and Latency; Systematic A/B Testing.
- **Machine Learning & Data Science:** Deep Learning (CNN, LSTM), Computer Vision (OpenCV), Time Series Forecasting; Predictive Modeling, Clustering, and Anomaly Detection.
- **Programming & Frameworks:** Python (Pandas, PyTorch, TensorFlow, Scikit-Learn, FastAPI), R, SQL; Async programming and RESTful API design.
- **Data Infrastructure & Cloud:** Azure (Databricks, Data Factory), GCP (BigQuery, Cloud Storage), AWS S3; PostgreSQL, Snowflake, Oracle, Teradata; Docker, Kubernetes, GitLab CI/CD.

PROJECTS

Nexus-Graph [demo]

- Developed an **open-source CLI tool and MCP server** that optimizes AI context retrieval, reducing token usage by up to **95%** through a surgical graph-based mapping system.
- Implemented **Abstract Syntax Tree (AST)** parsing using Tree-sitter to map code symbols (functions, classes, dependencies) into a high-performance SQLite-backed relational graph.
- Distributed via **NPM** and built a high-fidelity marketing site with an interactive Vis.js simulation dashboard to visualize complex code dependencies.

CostLine -AI Spend Intelligence & FinOps Platform [demo]

- Built an infrastructure-level proxy and intelligence platform that enforces hard budget limits (sub-10ms overhead) and tracks per-user/per-feature unit economics for LLM applications.
- Implemented automated optimization detection, analyzing traffic for RAG over-retrieval, history bloat, and model routing inefficiencies to provide actionable cost-saving recommendations.
- Designed the platform for enterprise self-hosting (Docker, Kubernetes, Terraform for AWS/GCP/Azure) to ensure zero customer data exfiltration for air-gapped environments.

localGPT (GitHub Open-Source Contribution)

- Developed and integrated a custom **JSON loader class** to streamline document ingestion and automated embedding generation using **Chroma vectorDB**.

AWARDS

- Third Prize Winner, Reinvent-The-Wheel 2.0 Hackathon (24-hour sprint) hosted by TORQATA & ATD; competed against 20+ professional data science teams. (Nov 2022)
- **Akhilesh Pothuri** and Pooja Malik. Knowledge graph augmented advanced learning models for commonsense reasoning. In *IOP Conference Series: Materials Science and Engineering*, volume 1022, page 012038. IOP Publishing, 2021[**Paper**]

EDUCATION

University of North Carolina at Charlotte

Master of Science in Data Science and Business Analytics

Shiv Nadar University

Bachelor of Technology in Computer Science and Engineering, Minor in Mathematics

Charlotte, North Carolina

August 2022 - December 2023

Greater Noida, India

July 2016 - July 2020